

Tarun Kathuria

CONTACT INFORMATION

E-mail: tarunkathuria@gmail.com
Homepage: tarunkathuria.github.io
Mobile: +1-510-963-0489

RESEARCH INTERESTS

(Non)-Convex Optimization, Stochastic Processes with applications to generative modelling, Randomized Linear Algebra, Reinforcement Learning with applications to Alignment of Language Models

GENERATIVE MODELS TRAINING EXPERIENCE

- Distributed Training of LMs with $O(1B)$ parameters on GPUs and TPUs with modifications to standard architectures (T5, UL2, GPT-2) to create non-autoregressive diffusion LMs
- Model Architecture Changes: Softmax attention replacements for OOD generalization tasks
- New Robust Self-Supervised Representation Learning (SSL) outperforming popular SSL methods on linear probing as well as some downstream tasks
- Modifications to Policy Gradient methods like PPO, GRPO to accomodate vector-valued reward models to improve reward hacking/regression for improved reasoning capabilities and alignment of LMs
- Incorporating PPO/GRPO variants for RL for diffusion/flow matching methods for image generation.
- Training Message Passing and Graph Transformer Neural Networks for use-cases in Google Maps

EMPLOYMENT

Google Research, Mountain View, CA
Research Scientist

Symbolica AI, San Carlos, CA
Machine Learning Researcher

- Worked on designing non-auto-regressive (AR) language models to match the performance of SOTA open-source AR language models
- Working on designing improved self-supervised embeddings for language and image generation tasks
- Worked on designing improvements to Proximal Policy Optimization which can use vector-valued reward models which allows for improved fine-grained alignment of language models

Adobe Research, San Jose, CA
Summer Research Intern

Mentor: Dr. Anup B. Rao

- Worked on designing new accelerated diffusion models for image generation
- Obtained significant speed-ups (33% lesser sampling time) on MNIST, CIFAR-10, ImageNet image generation tasks with competing FID scores to HuggingFace's DPM-Solver implementation

Microsoft Research, Bangalore, India
Research Fellow, Algorithms and Theory Group

Mentor: Dr. Amit Deshpande

- Worked on theoretical and applied problems in optimization and randomized numerical linear algebra
- Published some of this work in the context of Bayesian optimization (NIPS) and Fairness in Machine Learning (FATML, ICML) and theoretical computer science (APPROX-RANDOM 2017)

IBM Research, Bangalore, India
Summer Research Intern

Mentor: Dr. Indrajit Bhattacharya

- Worked on designing improvements to the Chinese Restaurant Process, a non-parameteric Bayesian method used for topic modelling of documents
- Designed an augmentation that can incorporate authorship of documents into the topic modelling
- Implemented the process using Hadoop/Map-Reduce for large scale inference and obtained improved results on a Twitter tweets dataset

EDUCATION

University of California, Berkeley
Ph.D.

- Major: Computer Science Minor: Mathematics
- Dissertation Title: Improved Interior Point Methods for Structured Combinatorial Problems

Indian Institute of Technology - Bombay, Mumbai, India
Bachelor of Technology (Honors)

- Major: Computer Science & Engineering Minor: Applied Statistics and Informatics

PUBLICATIONS

1. Tarun Kathuria, Sachin Kumar. [Leveraging Pretrained Language Models as Energy Functions for Glauber Dynamics Text Diffusion](#). Under submission ICLR 2026.

2. Ankit Garg, Tarun Kathuria, Nikhil Srivastava. [Scalar Poincare implies Matrix Poincare](#). *Electronic Communications in Probability* (26), 2021.
3. Tarun Kathuria. [A Potential Reduction Inspired Interior Point Method for Exact Max Flow in almost \$O\(m^{4/3}\)\$ Time](#). *IEEE Annual Symposium on Foundations of Computer Science (FOCS 2020)*
4. Haotian Jiang, Tarun Kathuria, Yin Tat Lee, Swati Padmanabhan, Zhao Song. [A Faster Interior Point Method for Semidefinite Programming](#). *IEEE Annual Symposium of Foundations of Computer Science (FOCS 2020)*
5. Yeshwanth Cherpanamjeri, Samuel B. Hopkins, Tarun Kathuria, Prasad Raghavendra, Nilesch Tripaneni. [Algorithms for heavy-tailed statistics: regression, covariance estimation, and beyond](#). *ACM-SIGACT Symposium on Theory of Computing (STOC 2020)*
6. Tarun Kathuria, Satyaki Mukherjee, Nikhil Srivastava. [On Concentration Inequalities for Random Matrix Products](#). Arxiv preprint. 2020
7. Tarun Kathuria [A Matrix Bernstein Inequality for Strong Rayleigh Distributions](#). Arxiv preprint. 2020
8. Elisa Celis, Vijay Keswani, Damnjan Straszak, Amit Deshpande, Tarun Kathuria, Nisheeth Vishnoi. [Fair and Diverse DPP-based data summarization](#). *International Conference on Machine Learning (ICML 2018)*
9. Tarun Kathuria, S. Sudarshan. [Efficient and Provable Multi-Query Optimization](#). *ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS 2017)*
10. Tarun Kathuria, Amit Deshpande, Pushmeet Kohli. [Batched Gaussian Process Bandit Optimization via Determinantal Point Processes](#). *Advances in Neural Information Processing Systems (NIPS 2016)*
11. L. Elisa Celis, Amit Deshpande, Tarun Kathuria, Nisheeth K. Vishnoi. [How to be Fair and Diverse?](#) *Workshop on Fairness, Accountability, and Transparency in Machine Learning (FATML 2016)* (oral)

EXTERNAL
REVIEWER

Conference on Learning Theory (COLT), International Conference on Machine Learning (ICML), Neural Information Processing Systems (NeurIPS), Symposium on Discrete Algorithms (SODA), Foundations of Computer Science (FOCS), Symposium on Theory of Computing (STOC)

TEACHING
EXPERIENCE

Graduate Student Instructor

Course: Convex Optimization (EECS127/227A)

Graduate Student Instructor

Course: Algorithms and Intractability (CS170)

Undergraduate Teaching Assistant

Course : Introduction to Numerical Analysis

Undergraduate Teaching Assistant

Course : Linear Algebra

Undergraduate Teaching Assistant

Course : Electricity & Magnetism

TECHNICAL
SKILLS

Programming
Machine Learning

Python, C++
JAX, Pytorch, Scikit-learn, Pandas, Numpy